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GitHub link:https://github.com/mumfali/practicals_webtech.git

Problem 2: Simple Shopping Cart

Problem statement: Develop a Shopping Cart application where users can view products, add products to a cart, remove products, and calculate the total price.

Requirements

1. Display product names, prices, and Add to Cart buttons.
2. Use JavaScript events and DOM manipulation to update the cart.
3. Store products as objects in arrays. Use filter() to remove items, map() to display
4. them, and reduce() to calculate the total.
5. Fetch product information from a public product API.
6. Create a basic React + TypeScript component for displaying products and managing
7. cart state.
8. Create a Git repository, commit the code, push it to GitHub, and demonstrate a
9. separate branch.

index.html :

```
<div class="box">
    <h1>Beauty Products</h1>
    <div id="product-list"></div>

    <h1>Shopping Cart</h1>
    <div id="cart-list"></div>
    <h2>Total: <span id="total-price">$0.00</span></h2>
</div>

<script src="script.js"></script>
```

// Requirement 3: Store beauty products as objects in arrays (with image links)

const products = [

{

id: 1,

title: "Matte Lipstick",

price: 18.99,

image: "https://images.unsplash.com/photo-1586495777744-4413f21062fa?auto=format&fit=crop&w=150&q=80"

},

{

id: 2,

title: "Hydrating Face Serum",

price: 34.50,

image: "https://images.unsplash.com/photo-1620916566398-39f1143ab7be?auto=format&fit=crop&w=150&q=80"

},

```
{
id: 3,
title: "Liquid Foundation",
price: 29.99,
image: "https://images.unsplash.com/photo-1631729371254-42c2892f0e6e?auto=format&fit=crop&w=150&q=80"
},
{
id: 4,
title: "Waterproof Mascara",
price: 15.00,
image: "https://media.istockphoto.com/id/485201410/photo/make-up-blue-eye-with-long-lashes-with-black-mascara.jpg?s=612x612&w=0&k=20&c=zqO8z2nofKbT_4DicftMaItbOZgKuAm7SpyilqfeUz4="
},
{
id: 5,
title: "Eyeshadow Palette",
price: 42.00,
image: "https://www.sugarcosmetics.com/cdn/shop/files/Blend-The-Rules-Eyeshadow-Palette-8.jpg?v=1743677313&width=1080"
}
];
let cart = [];
// Requirement 3: Use map() to display products with images
function displayProducts() {
const list = document.getElementById('product-list');
list.innerHTML = products.map(p =>          <div class="card">
            <div class="product-info">
                <img src="${p.image}" alt="${p.title}" class="product-img">
                <div>
                    <h3>${p.title}</h3>
                    <p>Price: $${p.price}</p>
                </div>
            </div>
            <button onclick="addToCart(${p.id})">Add to Cart</button>
        </div>
    ).join('');
}
// Requirement 2: JavaScript events & DOM manipulation to add items
function addToCart(id) {
const product = products.find(p => p.id === id);
const item = cart.find(i => i.id === id);
```

```
  if (item) {
      item.quantity += 1;
  } else {
      cart.push({ ...product, quantity: 1 });
  }
  updateCart();
```

```
}
// Decrease quantity or remove if count reaches 0
function decreaseQuantity(id) {
const item = cart.find(i => i.id === id);
if (item) {
if (item.quantity > 1) {
item.quantity -= 1;
} else {
removeFromCart(id);
return;
}
}
updateCart();
}
// Requirement 3: Use filter() to remove items completely
function removeFromCart(id) {
cart = cart.filter(i => i.id !== id);
updateCart();
}
function updateCart() {
const cartList = document.getElementById('cart-list');
```

```
 if (cart.length === 0) {
     cartList.innerHTML = '<p>Your cart is empty</p>';
 } else {
     // Requirement 3: Use map() to display cart items with options (+,
 -, remove)
     cartList.innerHTML = cart.map(i => `
         <div class="card">
             <div class="product-info">
                 <img src="${i.image}" alt="${i.title}" class="product-
 img">
                 <span>${i.title} (x${i.quantity}) - $${(i.price *
 i.quantity).toFixed(2)}</span>
             </div>
             <div>
                 <button onclick="decreaseQuantity(${i.id})">-</button>
                 <button onclick="addToCart(${i.id})">+</button>
                 <button class="danger"
 onclick="removeFromCart(${i.id})">Remove</button>
             </div>
         </div>
     `).join('');
 }

 // Requirement 3: Use reduce() to calculate the total price
 const total = cart.reduce((sum, i) => sum + (i.price * i.quantity), 0);
 document.getElementById('total-price').innerText =
 `$${total.toFixed(2)}`;

}
```

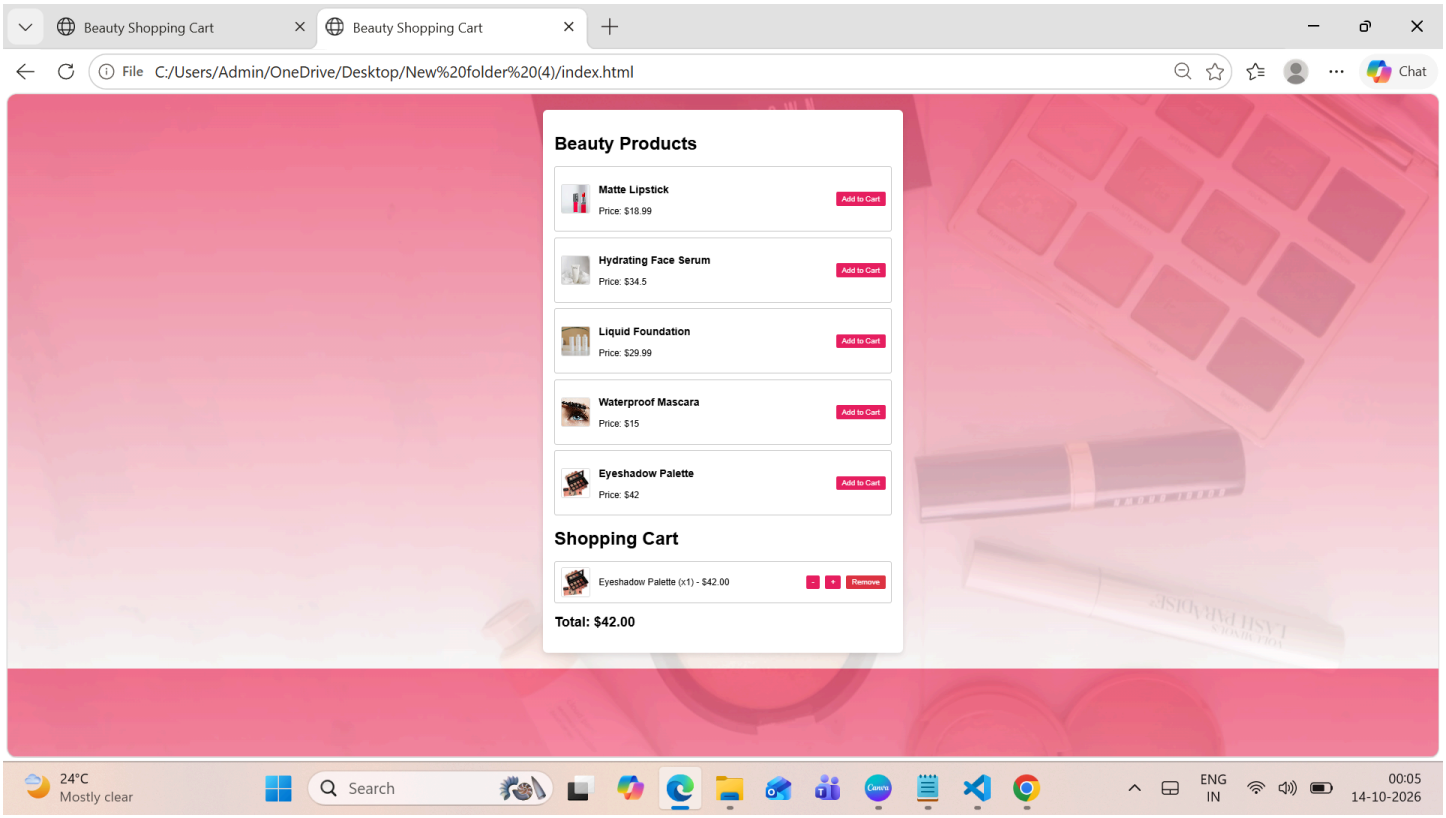
style.css

```
body {
font-family: Arial, sans-serif;
padding: 20px;
background: linear-gradient(rgba(245, 96, 131, 0.85), rgba(255, 255, 255, 0.85)),
url('https://images.unsplash.com/photo-1522335789203-aabd1fc54bc9?
auto=format&fit=crop&w=1600&q=80') center fixed;
background-size: cover;
}
.box {
max-width: 600px;
margin: auto;
background: white;
```

```
padding: 20px;
border-radius: 8px;
box-shadow: 0 4px 10px rgba(0,0,0,0.1);
}
.card {
border: 1px solid #ccc;
padding: 10px;
margin: 10px 0;
border-radius: 4px;
display: flex;
justify-content: space-between;
align-items: center;
}
/* Product info layout with image */
.product-info {
display: flex;
align-items: center;
gap: 15px;
}
.product-img {
width: 50px;
height: 50px;
object-fit: cover;
border-radius: 4px;
border: 1px solid #ddd;
}
button {
background: #e91e63;
color: white;
border: none;
padding: 5px 10px;
cursor: pointer;
border-radius: 3px;
margin-left: 5px;
}
button:hover {
background: #c2185b;
}
.danger {
background: #dc3545;
}
.danger:hover {
background: #c82333;
```

}

output:


Beauty Shopping Cart
Beauty Shopping Cart
File C:/Users/Admin/OneDrive/Desktop/New%20folder%20(4)/index.html
Chat
Beauty Products
Matte Lipstick
Price: $18.99
Add to Cart
Hydrating Face Serum
Price: $34.5
Add to Cart
Liquid Foundation
Price: $29.99
Add to Cart
Waterproof Mascara
Price: $15
Add to Cart
Eyeshadow Palette
Price: $42
Add to Cart
Shopping Cart
Eyeshadow Palette (x1) - $42.00
- + Remove
Total: $42.00
24°C
Mostly clear
Search
ENG
IN
00:05
14-10-2026